

CUSTOMER MILESTONE BRIEF

MS #04 - Graph Explorer and Neo4j Projection

Purpose, customer value, delivered capability, evidence, and claim boundaries

Status	Complete
Release	0.4.0
Date	2026-08-10
Audience	Customers, partners, executives, architects, delivery teams, and assurance reviewers

Executive summary

MS #4 - Graph Explorer and Neo4j Projection converts a specific part of the LIGHTYEAR modernization approach into a governed, reviewable capability. Makes the modernization graph understandable through visual exploration and a lossless external graph projection.

Why this matters to a customer

Makes the modernization graph understandable through visual exploration and a lossless external graph projection.

The practical result is a narrower, evidence-backed decision instead of a broad modernization claim. Commercial, architecture, delivery, security, and assurance teams can review the same capability, proof, and limits.

What the milestone delivers

- Added a locally executable LIGHTYEAR Graph Explorer with no third-party runtime dependencies.
- Added five curated exploration perspectives for workload, rule, JCL lineage, data contract, and discovered edge-case analysis.
- Added bounded search, node inspection, evidence display, pan, zoom, depth selection, and graph focus controls for the 10,000-node estate.
- Enforced implementer/verifier visibility across search, direct lookup, neighborhoods, and traces.
- Added a deterministic, lossless Neo4j CSV projection with graph identity and row-count receipts.

- Added Neo4j import guidance and explicit source-of-truth and private-evidence boundaries.
- Added HTTP, privacy, bounding, explorer, and Neo4j projection tests to the verification suite.

How it differs from earlier work

MS #4 builds on MS #3 (Evidence-Aware Knowledge Graph). The earlier milestone established its own bounded capability; this milestone adds graph explorer and neo4j projection while preserving the earlier evidence and its limits.

Earlier evidence is not automatically promoted into this milestone. A parser, simulator, local comparison, or generated artifact is not live platform equivalence or production readiness.

What a customer can do with it

A customer can use graph explorer and neo4j projection as a bounded decision and delivery capability, attached to approved sources, owners, policies, target architecture, acceptance criteria, and authorized evidence.

Unresolved semantic loss, policy decisions, unsupported behavior, and live-evidence requirements remain visible.

Evidence and acceptance posture

The repository capability is complete because its committed implementation and release record are present. Customer qualification remains claim-specific and evidence-class-specific.

Review exact source and artifact identities, distinguish development from authorized live evidence, inspect policy and loss classifications, confirm passed and blocked gates, and verify that later changes have not invalidated the evidence.

Boundaries and claims not made

- This milestone establishes only the bounded capabilities and evidence listed here; later capabilities are not retroactive.
- Production readiness and platform equivalence require the explicit evidence and policy gates applicable to the customer environment.

Source of record

- CHANGELOG.md release section(s): 0.4.0
- README.md product and operating guidance
- LIGHTYEAR-ROADMAP.md governing sequence and claim classifications
- docs/milestones/catalog.json metadata and docs/milestones/manifest.json artifact integrity

The underlying schemas, receipts, ledgers, tests, and policy decisions remain authoritative when greater detail is required.